

최적 제어 hw*4

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1) Prob 4-14

(p 181)

Find the optimal

$$J(x) = \int_0^{t_f} \left[\frac{\sqrt{1 + \dot{x}^2(t)}}{x(t)} \right] dt$$

$x(0) = 0 \Rightarrow x(t_f)$ must lie on the line $\theta(t) = t - 5$

$$E-L: \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) - \frac{\partial g}{\partial x} = 0$$

$$\frac{d}{dt} \left(\frac{1}{x} \cdot \frac{1}{2} (2\dot{x}) (1 + \dot{x}^2)^{-\frac{1}{2}} \right) - \left((1 + \dot{x}^2)^{-\frac{1}{2}} (-1) \dot{x}^2 \right) = 0$$

$$\frac{d}{dt} \left((\dot{x})^{-1} \dot{x} (1 + \dot{x}^2)^{-1/2} \right) - \left(-(\dot{x})^2 (1 + \dot{x}^2)^{-3/2} \right) = 0$$

$$(-1)(\dot{x})^{-2} \ddot{x} \dot{x} (1 + \dot{x}^2)^{-1/2} + (\dot{x})^{-1} \ddot{x} (1 + \dot{x}^2)^{-1/2} + (\dot{x})^{-1} \dot{x} \left(-\frac{1}{2} \right) 2\dot{x} \ddot{x} (1 + \dot{x}^2)^{-3/2} + (\dot{x})^{-2} (1 + \dot{x}^2)^{-3/2} = 0$$

$$\left[(1 + \dot{x}^2)^{-1/2} \left\{ -\frac{\ddot{x}^2}{\dot{x}^2} + \frac{\ddot{x}}{\dot{x}} \right\} + (1 + \dot{x}^2)^{-3/2} \left\{ -\frac{\ddot{x}^2}{\dot{x}} \ddot{x} \right\} + (1 + \dot{x}^2)^{-1/2} \left(\frac{1}{\dot{x}^2} \right) \right] = 0$$

$$\left[\frac{1}{\dot{x}^2} \left(-\ddot{x}^2 + \dot{x} \ddot{x} \right) (1 + \dot{x}^2) - \frac{\ddot{x}^2 \ddot{x}}{\dot{x}} + (1 + \dot{x}^2)^2 \frac{1}{\dot{x}^2} \right] \times \dot{x}^2 = 0$$

$$-\ddot{x}^2 - \ddot{x}^4 + \dot{x} \ddot{x}^3 + \dot{x}^2 \ddot{x}^2 \ddot{x} - \ddot{x}^2 \ddot{x}^2 \ddot{x} + (1 + 2\dot{x}^2 + \dot{x}^4) = 0$$

$$\therefore \ddot{x}^2 \ddot{x} + \dot{x}^2 \ddot{x} + 1 = 0$$

$$\Rightarrow \frac{d}{dt} \left[\dot{x}(t) \ddot{x}(t) \right] = -1 \Rightarrow \dot{x}(t) \ddot{x}(t) = -t + C$$

And

$$\frac{d}{dt} \left[\frac{1}{2} \dot{x}^2(t) \right] = \dot{x}(t) \ddot{x}(t) = -t + C$$

$$\therefore \frac{1}{2} \dot{x}^2(t) = -\frac{1}{2} t^2 + C_1 t + C_2$$

$$\dot{x}^2(t) = -t^2 + C_1 t + C_2 \quad \text{--- (1)}$$

$$B.C: \dot{x}^2(0) = 0 \rightarrow C_2 = 0$$

$$\dot{x}^2(t_f) = (t_f - 5)^2 \Rightarrow -t_f^2 + C_1 t_f = t_f^2 - 10t_f + 25 \quad \text{--- (2)}$$

$$\text{transversality condition: } g + \frac{\partial g}{\partial \dot{x}} \left[\dot{\theta} - \dot{x}^* \right] \Big|_{\dot{x}^*(t_f), \ddot{x}^*(t_f), t_f} = 0$$

$$\frac{(1 + \dot{x}^{*2})^{1/2}}{\dot{x}^*} + \frac{1}{x^*} \left(\frac{1}{2} \right) (2\dot{x}^*) (1 + \dot{x}^{*2})^{-1/2} (1 - \dot{x}^*) = 0$$

$$1 + \cancel{\dot{x}^{*2}} + \dot{x}^* - \dot{x}^{*2} = 0$$

$$1 + \dot{x}^*(t_f) = 0 \quad - (3)$$

$$(2) : 2t_f^2 - (c_1 + 10)t_f + 25 = 0 \quad - (4)$$

$$(1) : \dot{x}^*(t) = \sqrt{-t^2 + c_1 t}$$

$$\dot{x}^*(t) = (-2t + c_1) (-t^2 + c_1 t)^{-1/2}$$

$$(3) : \dot{x}^*(t_f) = -1$$

$$1 = (-2t_f + c_1)^2 (-t_f^2 + c_1 t_f)^{-1}$$

$$-t_f^2 + c_1 t_f = (-2t_f + c_1)^2 = 4t_f^2 - 4c_1 t_f + c_1^2$$

$$5t_f^2 - 5c_1 t_f + c_1^2 = 0 \quad - (5)$$

$$(4), (5) : t_f = 5 \pm \sqrt{\frac{25}{2}}$$

$$c_1 = 10$$

$$\therefore \dot{x}^{*2}(t) = -t^2 + 10t$$

2) Problem 4-16

Repeat Problem 4-14 with $x(0)=0$ & $x(t_f)$ lying on the circle $(t-9)^2 + x^2(t) = 9$

from 4-14

(E-L)

$$x^{*2}(t) = -t^2 + C_1 t + C_2$$

B.C: $x^{*2}(0)=0 \rightarrow C_2=0$

$$x^{*2}(t_f) = 9 - (t_f - 9)^2 \quad \text{--- (1)}$$

$$-2(t_f - 9) \cdot 2\dot{x}^* \dot{x}^*$$

transversality condition: $g(x, t_f) + \frac{\partial g}{\partial \dot{x}}(x, t_f)[\dot{x}(t_f) - \dot{x}^*(t_f)] = 0$

$$\frac{\sqrt{1 + \dot{x}^{*2}(t_f)}}{x^*(t_f)} + \frac{\dot{x}^*(t_f)}{x^*(t_f) \sqrt{1 + \dot{x}^{*2}(t_f)}} \times \left[\frac{-(t_f - 9)}{x^*(t_f)} - \dot{x}^*(t_f) \right] = 0$$

$$(1 + \dot{x}^{*2}(t_f)) + \dot{x}^*(t_f) \left[\frac{-(t_f - 9)}{x^*(t_f)} - \dot{x}^*(t_f) \right] = 0$$

$$x^*(t_f) - \dot{x}^*(t_f)(t_f - 9) = 0 \quad \text{--- (2)}$$

①: $2x^*(t_f)\dot{x}^*(t_f) = -2t + C_1$

②: $x^{*2}(t_f) - \dot{x}^* x^*(t_f)(t_f - 9) = 0$

$$9 - (t_f - 9)^2 - \left(-t_f + \frac{C_1}{2}\right)(t_f - 9) = 0$$

$$9 - (t_f^2 - 18t_f + 81) - (-t_f^2 + (9 + \frac{C_1}{2})t_f - \frac{9}{2}C_1) = 0$$

$$(18 - 9 - \frac{C_1}{2})t_f + (9 - 81 + \frac{9}{2}C_1) = 0$$

$$(9 - \frac{C_1}{2})t_f + (-72 + \frac{9}{2}C_1) = 0 \rightarrow \text{① or ② change}$$

$$\therefore C_1 = 8, \quad t_f = \frac{36}{5} = 7.2$$

$$\therefore x^{*2}(t) = -t^2 + 8t$$

3) Find $u(t)$, $x_1(t)$, $x_2(t)$ to extremize

$$J = \frac{1}{2} \int_0^1 x_1^2 + x_2^2 + u^2 dt$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 - x_2 + u$$

From book.

$$\begin{cases} \dot{x}^*(t) = \frac{\partial H}{\partial p}^*(x, t) \\ \dot{p}^*(t) = -\frac{\partial H}{\partial x}^*(x, t) \\ 0 = \frac{\partial H}{\partial u}^*(x, t) \end{cases}$$

$$H = \frac{1}{2}(x_1^2 + x_2^2 + u^2) + p_1(t) x_2(t) + p_2(t) \{-x_1(t) - x_2(t) + u(t)\}$$

$$\dot{p}_1^*(t) = -\frac{\partial H}{\partial x_1} = -(x_1^*(t) - p_2^*(t))$$

$$\dot{p}_2^*(t) = -\frac{\partial H}{\partial x_2} = -(x_2^*(t) + p_1^*(t) - p_2^*(t))$$

$$0 = \frac{\partial H}{\partial u} = u^*(t) + p_2^*(t) \rightarrow u^*(t) = -p_2^*(t)$$

$$\dot{x}_1^*(t) = x_2^*(t)$$

$$\dot{x}_2^*(t) = -x_1^*(t) - x_2^*(t) - p_2^*(t)$$

$$\begin{pmatrix} \dot{x}_1^*(t) \\ \dot{x}_2^*(t) \\ \dot{p}_1^*(t) \\ \dot{p}_2^*(t) \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & -1 & 0 & -1 \\ -1 & 0 & 0 & +1 \\ 0 & -1 & -1 & +1 \end{pmatrix}}_A \begin{pmatrix} x_1^*(t) \\ x_2^*(t) \\ p_1^*(t) \\ p_2^*(t) \end{pmatrix}$$

$$\begin{pmatrix} x_1^*(t) \\ x_2^*(t) \\ p_1^*(t) \\ p_2^*(t) \end{pmatrix} = e^{At} \begin{pmatrix} x_1^*(0) \\ x_2^*(0) \\ p_1^*(0) \\ p_2^*(0) \end{pmatrix}$$

$$t=1 \rightarrow e^{At} = \begin{pmatrix} 0.3858 & 0.6524 & 0.1663 & -0.4912 \\ -0.8187 & 0.4306 & 0.4912 & -0.9833 \\ -0.4843 & -0.8282 & 0.2533 & 1.8131 \\ 1.1601 & -0.8171 & -1.6468 & 2.3913 \end{pmatrix}$$

Prob 1a)

$$\vec{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{x}(1) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$0 = 0.5858 + 0.1663 p_1^*(0) - 0.4912 p_2^*(0)$$

$$0 = -0.8187 + 0.4912 p_1^*(0) - 0.9833 p_2^*(0)$$

$$\rightarrow p_1^*(0) = 11.1413, \quad p_2^*(0) = 5.10136$$

$$\rightarrow p_1^*(1) = 11.1509, \quad p_2^*(1) = -5.9411$$

$$\rightarrow u^*(0) = -p_2^*(0) = -5.10136$$

$$u^*(1) = -p_2^*(1) = 5.9411$$

MATLAB code & graph

Prob 2a)

$$\vec{x}(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \vec{x}(1) = \begin{pmatrix} \text{free} \\ 0 \end{pmatrix}$$

$$\frac{\partial h}{\partial \vec{x}}(*, t_f) - \vec{p}^*(t_f) = 0$$

$$x_1: \quad x_1^*(t_f) - p_1^*(t_f) = 0 \rightarrow x_1^*(1), p_1^*(1) = \text{free}$$

$$x_2: \quad x_2^*(t_f) - p_2^*(t_f) = 0 \rightarrow x_2^*(1) = p_2^*(1) = 0$$

$$x_2^*(1) = 0 = -0.8187 + 0.4306 + 0.4912 p_1^*(0) - 0.9833 p_2^*(0)$$

$$p_2^*(1) = 0 = 1.1607 - 0.8171 - 1.6468 p_1^*(0) + 2.3913 p_2^*(0)$$

$$\rightarrow p_1^*(0) = -1.38649, \quad p_2^*(0) = -1.09576$$

$$\rightarrow p_1^*(1) = -3.6506, \quad p_2^*(1) \approx 0 = x_2^*(1) \rightarrow \text{OK!}$$

$$\therefore u^*(0) = -p_2^*(0) = 1.09576$$

$$u^*(1) = -p_2^*(1) \approx 0$$

MATLAB code & graph

최적궤도제어 hw4

3) MATLAB code

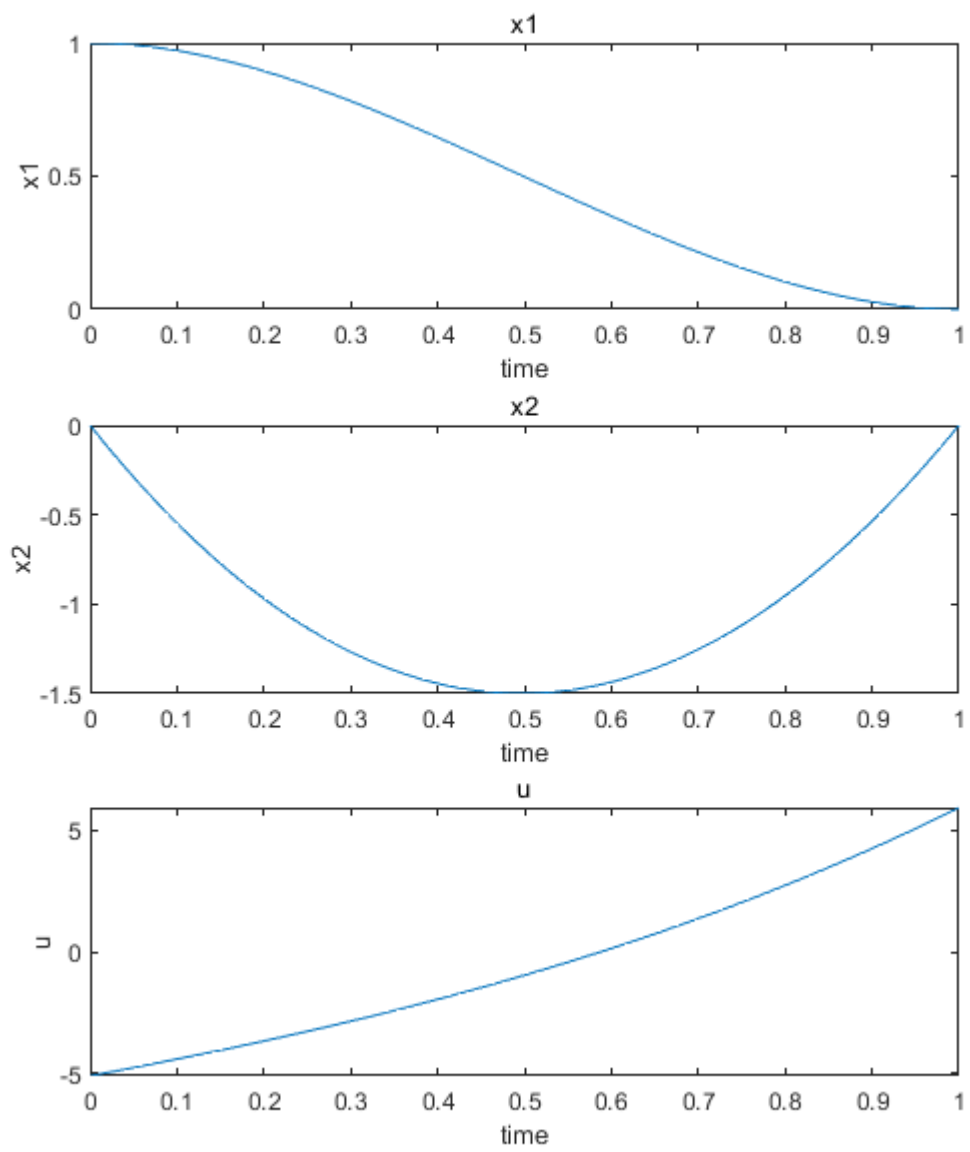
```
clc;
clear all;
tf=1;
A=[0,1,0,0;-1,-1,0,-1;-1,0,0,1;0,-1,-1,1];
x0=[1;0;11.7473;5.10736];
xb=[1;1;-1.38649;-1.09576];
expm(A*tf)*xb
time=linspace(0,tf,100);
x=zeros(4,100);
i=1;
for t=time
    x(:,i)=expm(A*t)*xb;
    i=i+1;
end

subplot(3,1,1);
plot(time,x(1,:));
title('x1')
xlabel('time')
ylabel('x1')

subplot(3,1,2);
plot(time,x(2,:));
title('x2')
xlabel('time')
ylabel('x2')

subplot(3,1,3);
plot(time,-x(4,:));
title('u')
xlabel('time')
ylabel('u')
```

Problem 1a) Graph



Problem 1b) Graph

